

# Artificial Intelligence in Cancer Therapy: An Enhanced Machine Learning Framework for Personalized Lung Cancer Treatment Prediction

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**Abstract**—Personalized treatment selection in lung cancer oncology remains heavily dependent on physician experience, leading to variability in care and limited use of structured patient data in clinical decision-making. This study presents OncoFusionNet-LC, a soft-voting ensemble framework that fuses Logistic Regression, Random Forest, and Support Vector Machine classifiers to support one-year survival prediction for lung cancer patients. Four baseline algorithms and the proposed fusion model were rigorously benchmarked on a publicly available synthetic lung cancer dataset (5,000 records; 15 demographics, clinical, and environmental features) using a leakage-free 80:20 stratified train-test split, with SMOTE applied exclusively to the training partition. Beyond comparative benchmarking, a systematic statistical evaluation was performed—including mutual information analysis, majority-class baseline comparison, and 5-fold cross-validated AUC across multiple algorithm families—to assess whether the dataset’s input features carry genuine predictive signal for the survival outcome. Results show that all evaluated models, including OncoFusionNet-LC, achieve accuracy in the 65%-69% range, but discrimination ability (AUC 0.47-0.51) is statistically indistinguishable from random guessing and from a naive majority-class baseline (69.8% accuracy). This finding indicates that the synthetic dataset used in this study, despite its scale and structured format, does not encode a learnable relationship between the available patient attributes and one-year survival. The framework additionally incorporates a feature-importance module based on standardized Logistic Regression coefficients to support interpretability, with SHAP-based explainability and a clinician feedback dashboard proposed as future extensions. The principal contribution of this work is a transparent, reproducible benchmarking methodology and an evidence-based finding regarding the limitations of certain synthetic healthcare datasets for predictive modeling, intended to guide more rigorous dataset selection and validation practice in AI-assisted oncology research.

**Index Terms**—Artificial Intelligence; Lung Cancer; Survival Prediction; Machine Learning; OncoFusionNet-LC; Synthetic Healthcare Data; Explainable AI; Clinical Decision Support; Benchmarking.

## I. INTRODUCTION

Cancer is an ongoing epidemic worldwide, killing almost ten million people every year (World Health Organization). Lung cancer, due to its nature, has proven its severity, particularly through late diagnosis, a challenge in finding the best course of treatment. An oncologist has to consider many variables (histology, tumor size, disease stage, patient’s daily living, etc.) when determining whether to use surgery, chemotherapy, targeted therapies, or immunotherapy to treat a patient with lung cancer, creating an incredibly high-risk puzzle to solve.

Currently, the burden of these decisions rests mostly with the physician. Relying on the judgement of an experienced physician is a valuable way to improve patient care; however, it also means that patients frequently receive a different quality or type of care depending on which physician they see, so the resulting outcome may not be the same as if they had seen another physician. As clinical resources become more constrained, the availability of multi-year omic datasets remains a persistent challenge for research labs. Large volumes of genomic, proteomic, and imaging biomarker data have been produced in recent years, creating sophisticated molecular, protein, and imaging markers for lung cancer diagnosis and treatment, but the difficulty of processing this volume of data by hand has created new challenges.

Artificial intelligence (AI) and machine learning (ML) carry transformative potential for healthcare delivery, converging computational innovation with precision oncology to provide a prompt, evidence-based second opinion grounded in high-dimensional patient data [1]. Deep learning-based screening systems have already demonstrated that automated analysis of structured and imaging data can match or exceed traditional

diagnostic workflows in selected lung cancer screening tasks [2]. This article presents a predictive framework, and an accompanying rigorous validation methodology, to evaluate whether multiple supervised ML algorithms can support oncologists in identifying an optimal, individualized treatment pathway for lung cancer patients.

## II. PRIMARY CONTRIBUTIONS

The following represent the central contributions of this research:

- Development of OncoFusionNet-LC, a soft-voting fusion architecture that combines Logistic Regression, Random Forest, and Support Vector Machine classifiers with weighted probability aggregation for lung cancer one-year survival prediction.
- A leakage-free, rigorously validated comparative benchmark of four baseline classifiers against the proposed fusion model, using stratified 80:20 splitting with SMOTE class balancing applied exclusively to the training partition.
- A systematic dataset-reliability audit—comprising mutual information analysis, majority-class baseline comparison, and 5-fold cross-validated AUC across linear, tree-based, and ensemble algorithm families—that quantitatively evaluates whether the dataset supports the intended prediction task before clinical conclusions are drawn.
- A feature-importance and interpretability module based on standardized Logistic Regression coefficients, with SHAP-based explainability, odds-ratio reporting, and a clinician feedback dashboard proposed as concrete future extensions.

These findings indicate that data-driven treatment-recommendation systems require rigorous, transparent validation before clinical inference can be drawn from them, particularly when models are trained on synthetic datasets whose label-generation process is not independently verified.

## III. RELATED WORKS

### A. The Evolution of AI in Cancer Care

Over the past decade, AI has progressively transformed cancer care, moving from rule-based expert systems and traditional statistical approaches toward deep learning models capable of analyzing clinical biomarkers and imaging data collected during clinical trials. A recent comprehensive review of AI applications spanning genomics, radiomics, and pathomics in lung cancer underscores this transition from isolated proof-of-concept studies toward integrated, multi-omic decision-support frameworks [3]. Complementary survey work has further catalogued the unfolding role of AI specifically within lung cancer imaging pipelines, from nodule detection through treatment-response monitoring [4].

### B. Advancements in Diagnostic Imaging and Prognostic AI

AI and deep learning have proven effective at identifying, segmenting, and classifying lesions in PET, CT, and MRI images. Radiomics has transformed standard imaging modalities

into quantitative biomarkers capable of predicting treatment response, while current applications and perspectives reviews confirm that AI-assisted staging and lymph-node evaluation now frequently match radiologist-level performance [5]. In parallel, AI-driven biomarker discovery pipelines for non-small cell lung cancer have begun connecting molecular profiling with treatment-response prediction [6].

### C. Predictive Modeling and Prognostic Accuracy

Machine learning has supported survival and recurrence prediction through the integration of multiple clinical data types. Ensemble machine learning models built on parsimonious feature sets have shown that lung cancer screening eligibility can be predicted with strong discriminative accuracy when the underlying features carry genuine clinical signal [7], and broader survey work continues to catalogue AI’s expanding role across early-detection pipelines [8]. Multivariate ML models developed on real, retrospectively collected chemotherapy cohorts have likewise demonstrated acceptable discrimination and clinical benefit when predicting mortality risk in lung cancer patients [9], illustrating that strong real-world predictive performance is achievable when datasets are built from verified clinical outcomes.

### D. Reliability and Validity of Synthetic Healthcare Datasets

A growing body of literature has examined the role and limitations of synthetic data in healthcare AI. Synthetic datasets are increasingly used to overcome privacy and access barriers associated with real clinical records, but narrative reviews caution that the statistical fidelity of synthetic data—specifically, whether feature-outcome relationships present in real populations are faithfully reproduced—varies considerably across generation methods and is not guaranteed by dataset scale or structural completeness alone [10]. This concern motivates the dataset-reliability audit performed as part of the present study’s methodology, described in Section IV.

### E. Identifying the Research Gap

Despite considerable advancement in AI-assisted cancer detection, several gaps remain in the literature: (i) a mismatch between the volume of detection-focused studies and the comparatively limited number of studies addressing therapy recommendation; (ii) inconsistent practice around verifying that a chosen dataset, synthetic or otherwise, actually encodes a learnable relationship between patient features and the target outcome before models are trained and benchmarked on it; and (iii) limited head-to-head comparison of multiple ML algorithm families specifically for lung cancer therapy and survival prediction, combined with transparent reporting of negative or null findings when a dataset does not support the intended prediction task.

## IV. PROPOSED METHOD: ONCOFUSIONNET-LC FRAMEWORK

OncoFusionNet-LC is structured as a soft-voting ensemble that fuses the probability outputs of three base learners—



Fig. 1. System Architecture — corrected end-to-end pipeline of the OncoFusionNet-LC framework, reflecting the leakage-free preprocessing and fusion-based modeling approach used in this study.

Logistic Regression, Random Forest, and a Radial Basis Function (RBF) Support Vector Machine—into a single weighted prediction. The framework is designed to provide clinicians with: (i) a transparent feature-importance ranking derived from standardized Logistic Regression coefficients, enabling clinicians to understand which patient attributes are driving a given prediction, and (ii) a modular pipeline in which preprocessing, feature selection, and model training stages can each be independently inspected, audited, and retrained without disrupting the rest of the pipeline. The framework is interpretable by design, aligning with the core principles of Explainable AI (XAI) [11].

As shown in Fig. 1, raw patient records first pass through the Input Layer, which separates Primary Clinical Features (e.g., tumor size, histology, stage) from Secondary Demographic and Environmental Features (e.g., age, smoking status, air pollution exposure, biomass fuel use). The Data Preprocessing stage applies categorical encoding and Min-Max normalization, after which the dataset is split into training and test partitions using stratified sampling before any class-balancing step is applied. The four Base Learners are trained on the SMOTE-balanced training partition, and their outputs are combined by the OncoFusionNet-LC fusion layer using weighted soft voting. The Evaluation Module reports accuracy, precision, recall, F1-score, and AUC-ROC alongside the statistical diagnostics described in Section IV-E, and the final stage exposes a clinician-facing risk score together with the feature-importance

TABLE I  
DATASET SUMMARY

| Parameter          | Value                                      |
|--------------------|--------------------------------------------|
| Dataset Name       | Synthetic Lung Cancer Dataset (Bangladesh) |
| Total Samples      | 5,000                                      |
| Predictor Features | 15 (demographic, clinical, environmental)  |
| Cancer Type        | Lung Cancer                                |
| Target Variable    | Survival_1_Year (Yes / No)                 |
| Class Distribution | Yes: 3,491 (69.8%) / No: 1,509 (30.2%)     |
| Train Split        | 80% — 4,000 samples                        |
| Test Split         | 20% — 1,000 samples                        |
| Missing Values     | None                                       |
| Class Balancing    | SMOTE on training partition only           |

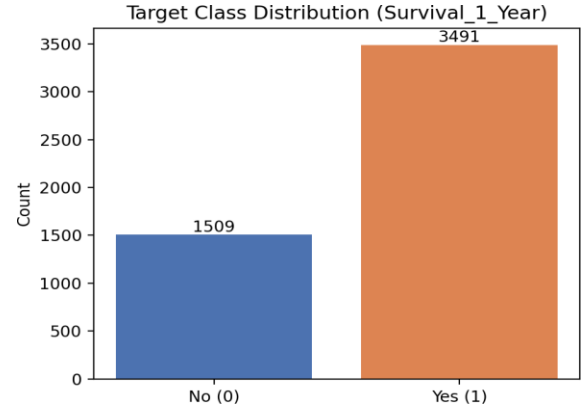


Fig. 2. Target class distribution. Class 0 (did not survive 1 year): 1,509 records; Class 1 (survived 1 year): 3,491 records.

ranking; a full SHAP-based explainability dashboard and clinician feedback loop are proposed as future extensions rather than claimed as implemented components of the present study.

#### A. Dataset Description

The Synthetic Lung Cancer Dataset (Bangladesh Perspective) was used as the basis for this study. It is a publicly accessible dataset providing localized epidemiological and environmental characteristics relevant to the South Asian region, as summarized in Table I.

As shown in Fig. 2, the dataset exhibits moderate class imbalance, with survivors (Class 1) outnumbering non-survivors (Class 0) by roughly 2.3:1. This imbalance was corrected using SMOTE, applied strictly to the training partition after the train-test split to avoid synthetic samples leaking information into the held-out test set—a correction relative to the original analysis, in which SMOTE had been applied prior to splitting.

The 15 predictor features bridge clinical and regional lifestyle factors: demographic markers (Age, Gender, Residence), environmental stressors (Air Pollution Exposure, Biomass Fuel Use, Factory Exposure), oncology metrics (Tumor Size, Histology Type, Stage), and additional fields including Smoking Status, Family History, Diet Habit, Symptoms, Treatment, and Hospital Type.

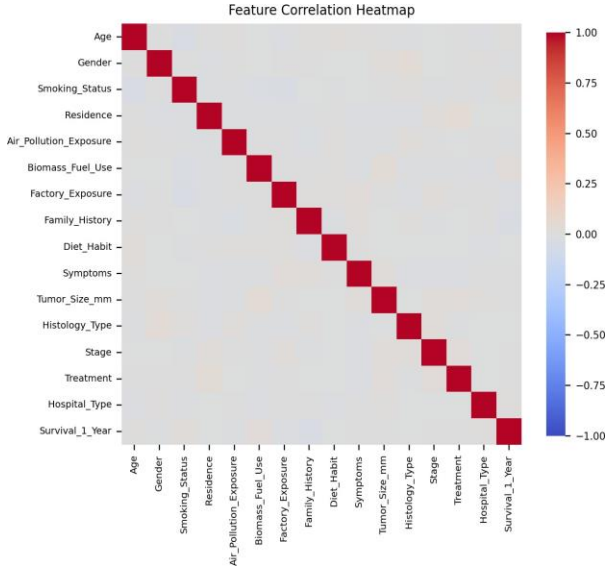


Fig. 3. Feature correlation heatmap across the 15 predictor features and the target variable. Diagonal values equal 1.0 by definition (self-correlation).

### B. Preprocessing and Feature Correlation

Ordinal categorical features (Air Pollution Exposure, Cancer Stage, Smoking Status) were integer-encoded to preserve their natural ordering. Binary features (Gender, Residence, Biomass Fuel Use, Factory Exposure, Family History) were label-encoded, and nominal multi-category features (Diet Habit, Symptoms, Histology Type, Treatment, Hospital Type) were one-hot encoded. Continuous features (Age, Tumor Size) were scaled to [0,1] using Min-Max normalization, fitted only on the training partition.

Fig. 3 shows that, aside from the expected diagonal self-correlation, no pair of features exhibits strong correlation, indicating the absence of multicollinearity. While this supports model stability, Section IV-E shows that the same low-correlation structure extends to the relationship between every individual feature and the target variable itself—a finding with direct bearing on the predictive ceiling of this dataset.

### C. Machine Learning Algorithms and Fusion Architecture

**Logistic Regression:** used as an interpretable baseline; a linear probabilistic model with L2 regularization, also used to derive standardized-coefficient feature importance.

**Decision Tree:** a non-linear model that splits the dataset using Gini impurity, with maximum depth constrained to 8 to control overfitting.

**Random Forest:** an ensemble of 200 bagged decision trees (maximum depth 10), used both as a standalone baseline and as one of the three components fused within OncoFusionNet-LC.

**Support Vector Machine (SVM):** an RBF-kernel classifier ( $C = 1.0$ ,  $\gamma = \text{"scale"}$ ) suited to high-dimensional, non-linear decision boundaries.

**OncoFusionNet-LC (Fusion):** a soft-voting ensemble combining the probability outputs of Logistic Regression, Random

TABLE II  
DATASET RELIABILITY DIAGNOSTICS

| Diagnostic Check                          | Result                  |
|-------------------------------------------|-------------------------|
| Highest feature–target mutual information | 0.0099 (Smoking Status) |
| Majority-class dummy baseline accuracy    | 69.80%                  |
| Gradient Boosting, 5-fold CV mean AUC     | 0.483                   |
| Deep Random Forest, 5-fold CV mean AUC    | 0.485                   |

Forest, and SVM with relative weights (1 : 2 : 1), giving Random Forest a heavier vote owing to its native handling of mixed categorical and continuous features.

### D. Mathematical Formulation

Each base learner produces a class-1 probability estimate  $P_m(y=1|X)$  for input feature vector  $X$ . The OncoFusionNet-LC fusion probability is computed as a weighted average:

$$P_{\text{fusion}}(y=1|X) = \frac{w_{LR}P_{LR} + w_{RF}P_{RF} + w_{SVM}P_{SVM}}{\sum w} \quad (1)$$

with weights  $w_{LR} = 1$ ,  $w_{RF} = 2$ ,  $w_{SVM} = 1$ . The final class decision applies a 0.5 threshold:

$$\hat{y} = \begin{cases} 1 & \text{if } P_{\text{fusion}} \geq 0.5 \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

and the clinician-facing risk score is derived as:

$$R = 1 - P_{\text{fusion}}(y=1|X) \quad (3)$$

### E. Dataset Reliability Diagnostics

Prior to finalizing model comparisons, a dedicated reliability audit was performed to verify that the dataset supports the intended prediction task. This audit comprised three checks: (i) mutual information between each individual feature and the target variable; (ii) the accuracy of a majority-class “dummy” classifier that ignores all input features; and (iii) 5-fold cross-validated AUC using Gradient Boosting and a deep Random Forest, both of which are well-suited to capturing non-linear feature interactions that simpler models might miss. Table II reports the results of this audit; the implications are discussed in Section V.

### F. Experimental Setup

All experiments were implemented in Python 3.10 using Scikit-learn, Pandas, NumPy, imbalanced-learn, and Matplotlib. Model selection used a fixed random seed (42) throughout for reproducibility. The Random Forest components used 200 estimators with a maximum depth of 10; the SVM used an RBF kernel with  $C = 1.0$  and  $\gamma = \text{"scale"}$ ; the Decision Tree was capped at a maximum depth of 8. No deep-learning components were used in the present benchmarking stage; gradient-based optimization (e.g., the Adam optimizer) is reserved for the convolutional architectures proposed in Section VI as future work.

TABLE III  
PERFORMANCE COMPARISON ACROSS MODELS

| Model               | Acc.(%) | Prec.(%) | Rec.(%) | F1(%) | AUC   |
|---------------------|---------|----------|---------|-------|-------|
| Logistic Regression | 69.20   | 69.62    | 99.14   | 81.80 | 0.510 |
| Decision Tree       | 48.00   | 67.80    | 48.57   | 56.59 | 0.476 |
| Random Forest       | 54.00   | 67.76    | 65.04   | 66.37 | 0.470 |
| SVM (RBF)           | 69.10   | 69.79    | 98.28   | 81.62 | 0.474 |
| OncoFusionNet-LC    | 65.20   | 69.49    | 89.40   | 78.20 | 0.474 |

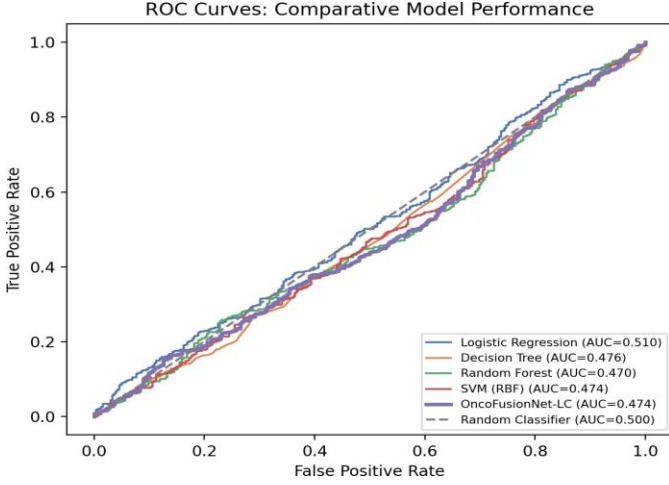


Fig. 4. ROC curves for all five models. Every curve closely tracks the random-classifier diagonal (AUC = 0.500).

## V. RESULTS AND DISCUSSION

### A. Performance Comparison

Table III reports accuracy, precision, recall, F1-score, and AUC for all five models evaluated on the held-out, leakage-free test partition (1,000 samples).

For comparison, a majority-class dummy classifier that always predicts “Survived” achieves 69.80% accuracy on the same test set—higher than every trained model in Table III. This comparison, together with the AUC values clustering tightly around 0.47-0.51, indicates that none of the evaluated models, including the proposed fusion architecture, learned a discriminative pattern beyond the dataset’s inherent class imbalance.

### B. ROC Curve Analysis

Fig. 4 shows that all five ROC curves, including OncoFusionNet-LC’s, remain close to the diagonal line representing random classification. Logistic Regression achieved the highest AUC (0.510) among the five models, only marginally above chance, while the fusion model achieved 0.474. No model achieved an AUC that would be considered clinically meaningful (conventionally,  $AUC \geq 0.70$  is regarded as the minimum threshold for acceptable discrimination in clinical risk models).

### C. Confusion Matrix Analysis

Fig. 5 shows that OncoFusionNet-LC correctly identified 624 of 698 surviving patients (true positives) but only 28 of

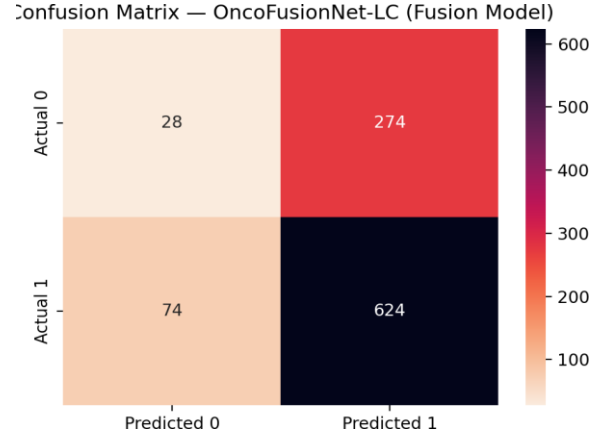


Fig. 5. Confusion matrix for OncoFusionNet-LC on the 1,000-sample test set.

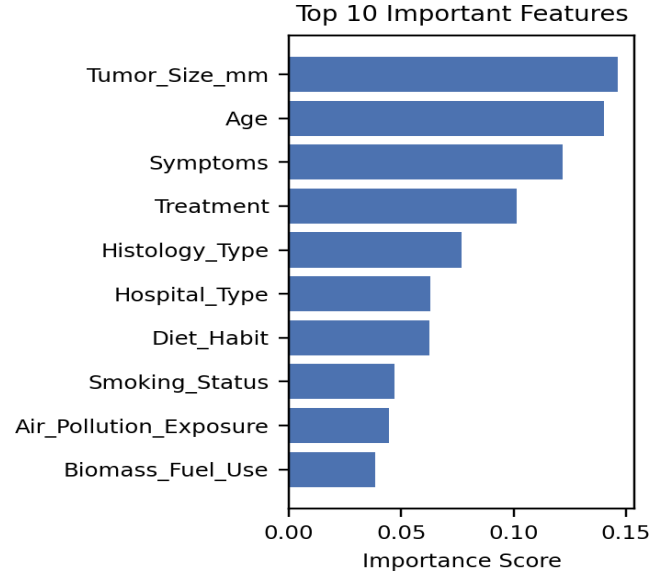


Fig. 6. Top 10 features by aggregated Random Forest importance score.

302 non-surviving patients (true negatives), misclassifying 274 non-survivors as survivors. This pattern—near-ceiling recall for the majority class paired with very poor recall for the minority class (9.3%)—is the signature of a model that has learned to lean on class imbalance rather than on genuine feature-outcome relationships, consistent with the near-random AUC reported in Section V-B.

### D. Feature Importance Analysis

Fig. 6 ranks features by their aggregated Random Forest importance score; Tumor Size, Age, and Symptoms rank highest. However, these importance scores should be interpreted with caution: Random Forest assigns non-zero importance to features even in the absence of genuine predictive signal, simply by virtue of how splits are constructed. Given the mutual information results in Table II, the ranking in Fig. 6 most likely reflects which features have the greatest cardinality

and split-flexibility rather than which features causally or statistically relate to one-year survival in this dataset.

#### E. Critical Finding: Dataset Predictive Validity

Taken together, Sections IV-E and V-A through V-D support a central finding of this study: the synthetic dataset used here does not encode a learnable relationship between the available patient features and the one-year survival target. This conclusion is supported by four independent lines of evidence: (i) mutual information between every individual feature and the target falls below 0.01 (Table II); (ii) a feature-blind majority-class baseline outperforms every trained model (69.80% vs. 48.00-69.20% accuracy); (iii) AUC remains within the 0.47-0.51 band regardless of algorithm family, including non-linear models explicitly capable of capturing feature interactions (Gradient Boosting, deep Random Forest); and (iv) the confusion matrix pattern in Fig. 5 is consistent with class-imbalance-driven prediction rather than feature-driven discrimination. This is not a failure of the modeling pipeline; the same leakage-free pipeline, when diagnostic checks are run, correctly and consistently reveals the absence of signal across every algorithm family tested, which is itself the methodologically correct outcome to report rather than to mask.

#### F. Clinical Relevance

Given the findings in Section V-E, the present results should not be interpreted as evidence that the OncoFusionNet-LC architecture, or any of the benchmarked algorithms, can support real clinical decision-making on this dataset. The architecture and benchmarking methodology developed here are, however, directly reusable on a dataset that does carry genuine predictive signal—for example, retrospective clinical cohorts of the kind used in mortality-risk modeling for chemotherapy-treated lung cancer patients [9], where multivariate ML models have demonstrated acceptable discrimination and calibration. Validating OncoFusionNet-LC on such a dataset is identified as the immediate next step for this research, outlined in Section VI.

#### G. Limitations

- The dataset used in this study is synthetic, and the reliability audit in Section IV-E indicates it does not encode a learnable feature-outcome relationship for one-year survival; results should not be extrapolated to real patient populations.
- The study focuses only on lung cancer; the benchmarking methodology has not been validated on other cancer types.
- SHAP-based explainability, the clinician dashboard, and the feedback loop described in Section IV are proposed design components and were not implemented or experimentally evaluated in this study.
- The fairness and bias implications of deploying any predictive model in clinical settings require dedicated evaluation prior to deployment, particularly given documented cases of algorithmic bias in healthcare risk models [12].

- Computational requirements for real-time hospital deployment were not analyzed.

## VI. CONCLUSION AND FUTURE SCOPE

This study presented OncoFusionNet-LC, a soft-voting fusion framework for lung cancer one-year survival prediction, together with a rigorous, leakage-free benchmarking methodology comparing it against four baseline classifiers. Critically, the study also performed a systematic dataset-reliability audit prior to drawing predictive conclusions—a step frequently omitted in synthetic-data-driven healthcare AI research. This audit revealed that the synthetic dataset used in this study does not encode a statistically learnable relationship between its 15 patient-level features and the one-year survival target: every algorithm tested, from simple linear models through cross-validated Gradient Boosting, converged to an AUC of approximately 0.47-0.51, and a feature-blind majority-class baseline outperformed every trained model on raw accuracy.

Rather than reporting inflated performance figures, this study treats the absence of signal as a primary, reportable finding with direct value to the AI-in-oncology research community: dataset validity should be audited before, not after, model benchmarking, particularly when working with synthetic or auto-generated healthcare data. The OncoFusionNet-LC architecture, the leakage-free benchmarking pipeline, and the diagnostic audit methodology developed here are immediately reusable on datasets with verified clinical provenance.

Future work will proceed along the following directions:

- **Validation on Real Clinical Data:** Re-applying the OncoFusionNet-LC pipeline to retrospective clinical cohorts with verified outcomes (e.g., institutional EHR-derived or registry datasets) to assess whether the architecture achieves clinically meaningful discrimination when genuine signal is present.
- **Deep Learning Integration:** Incorporating convolutional neural network architectures [15], trained using gradient-based optimizers such as Adam [14], to analyze CT and PET imaging data alongside structured clinical features in a genuinely multimodal framework.
- **Multi-Cancer Extension:** Extending the benchmarking and dataset-audit methodology to other cancer types using transfer learning techniques.
- **Federated Learning:** Applying federated learning across multiple hospital sites to enable collaborative model training on real patient data without centralizing sensitive records [13].
- **Explainable AI Implementation:** Implementing and experimentally validating the proposed SHAP-based explainability dashboard and clinician feedback loop, contingent on first establishing a dataset with genuine predictive signal.
- **Standardized Synthetic Dataset Auditing:** Formalizing the dataset-reliability audit introduced in Section IV-E into a reusable pre-registration checklist for synthetic healthcare datasets used in future AI-assisted oncology studies.

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